

11/30/2011

Dear Vazir,

Let $\mathcal{C}$ be a Tannakian category over an algebraically closed field k . Your question amounts to: does $\mathcal{C}$ always have a fiber functor over k . I believe it is true. The following argument is presumably too pedestrian, hiding which "compactness arguments" are really relevant.

Let A be the set of strictly full subcategories of $\mathcal{C}$, stable by $\otimes$, subquotients and duals. Let A_{fg} be the set of those generated by finitely many objects, using the same operations. For $\alpha \in A$, I note $\mathcal{C}_\alpha$ the corresponding subcategory (as defined: $\mathcal{C}_\alpha = \alpha$). The set A is ordered by inclusion. I assume we already know the existence and unicity up to isomorphism of fiber functors for the $\mathcal{C}_\alpha, \alpha \in A_{\text{fg}}$. If $\mathcal{C}_\alpha \subset \mathcal{C}_\beta$, it makes sense to say that a fiber functor ω_β on $\mathcal{C}_\beta$ extends (on the nose) a fiber functor ω_α on $\mathcal{C}_\alpha$. If an extension up to isomorphism exists, an actual extension exists too.

Let us order the set of $(\alpha, \omega_\alpha) : \alpha \in A, \omega_\alpha$ fiber functor on $\mathcal{C}_\alpha$, by " ω_β extends ω_α ". To avoid set theoretical difficulties, one could consider only the fiber functors ω_α such that the $\omega_\alpha(x)$ take values in the set of vector spaces k^n , $n \in \mathbb{N}$.

The ordered set of (α, ω_α) is inductive (Bourbaki
Ems Ch 3 §2.4): If I is a totally ordered subset,
the "union" of the (α, ω_α) is a $\mathcal{C}_0$ in A ($= \bigcup_{(\alpha, \omega_\alpha) \in I} \mathcal{C}_\alpha$)
with the fiber functor ω_0 characterized by
 $\omega_0|_{\mathcal{C}_\alpha} = \omega_\alpha$. This "union" majorizes I . By Bourb
loc cit Th 2, the ordered set of the (α, ω_α) has a
maximal element: $(\mathcal{C}_1, \omega_1)$. To prove that $\mathcal{C}_1 = \mathcal{C}$,
it suffices to prove the following

Lemma 1 Let $\mathcal{C}'$ be in A and $\mathcal{C}''$ be in A_f . Let
 $\langle \mathcal{C}', \mathcal{C}'' \rangle$ be generated by $\mathcal{C}'$ and $\mathcal{C}''$: it is in A .

Then, any fiber functor ω' on $\mathcal{C}'$ can be extended (or the
nose, or up to isomorphism, this amounts to the same)
to $\langle \mathcal{C}', \mathcal{C}'' \rangle$

I first prove

Lemma 2 Suppose $\mathcal{C}'$ is in $\mathcal{E}A_f$ too. "Restriction" is ~~an~~ then a
equivalence of categories

(fiber functors on $\langle \mathcal{C}', \mathcal{C}'' \rangle$) $\longrightarrow$ (triples of
a fiber functor ω' on $\mathcal{C}'$, ω'' on $\mathcal{C}''$ and an isomorphism τ
of the restrictions of ω' and ω'' to $\mathcal{C}' \cap \mathcal{C}''$ (also in A_f)).

We may assume that $\langle \mathcal{C}', \mathcal{C}'' \rangle$ is the category of
representations of a group G , and that for invariant subgroups
 A and B , $\mathcal{C}'$ (resp $\mathcal{C}''$) is the subcategory of representations
where A (resp B) acts trivially. That they generate $\langle \mathcal{C}', \mathcal{C}'' \rangle$
mean that $A \cap B = \{1\}$. The intersection $\mathcal{C}' \cap \mathcal{C}''$ is the category

of representations on which AB acts trivially

The triples $(\omega', \omega'', \tau)$ are all isomorphic :
as all ω' (resp all ω'') are isomorphic, it suffices
to see that $(\omega', \omega'', \tau_1)$ and $(\omega', \omega'', \tau_2)$ are isomorphic.
Indeed τ_1 and τ_2 differ by an automorphism of $\omega' \mid \tau' \cap \tau''$,
and such an automorphism lifts to an automorphism of ω' :

$$G/A(\mathbb{L}) \rightarrow G/AB(\mathbb{L}) \quad \text{is onto.}$$

We hence have here categories with just one
isomorphism class of objects, and the question is to
compare automorphism groups. We need to check

$$G \xrightarrow{\sim} \{ (g', g'') \in G/A, G/B \mid g' \text{ and } g'' \text{ have same image in } G/AB \}$$

$$\begin{array}{ccc} G & \longrightarrow & G/A \\ \downarrow & \square & \downarrow \\ G/B & \longrightarrow & G/AB \end{array}$$

Injectivity of this morphism of groups amounts to $A \cap B = \{e\}$,

Surjectivity : if $g' \in G/A, g'' \in G/B$ lift $\bar{g}', \bar{g}''$ left g', g'' ,
 $\bar{g}'' = \bar{g}' \cdot a \cdot b$ and $\bar{g}'' \cdot b = \bar{g}' \cdot a \in G$ maps to (g', g'') .

Lemma 3 . Same as Lemma 2, but τ' only assumed to be in A

Let B be the set of τ_a ($a \in A_f$) contained in τ' .
One has $\langle \tau', \tau'' \rangle = \bigcup_{\beta \in B} \langle \tau_\beta, \tau'' \rangle$. One has
equivalences

(fiber functors on τ) $\xrightarrow{\sim}$ (fiber functors^{exp} on the $\tau_p \otimes \tau'$,
plus a compatible system of isomorphisms $\omega_p \mid \tau_\beta \xrightarrow{\sim} \omega_\beta$
for $\tau_\gamma \subset \tau_\beta$; compatible : condition for $\tau_\delta \subset \tau_\gamma \subset \tau_\beta$)

Same for fiber functors on the $\langle \mathcal{C}', \mathcal{C}'' \rangle = \bigcup \langle \mathcal{C}_\beta, \mathcal{C}'' \rangle$.

The $\mathcal{C}_\beta \cap \mathcal{C}''$ have a ~~smallest~~^{largest} element: they if $\mathcal{C}'' = \text{Rep}(G'')$, they correspond to invariant subgroups of G'' , subgroups are closed subspaces, and one uses the noetherian property. If β_0 is such that $\mathcal{C}_{\beta_0} \cap \mathcal{C}''$ is the largest $\mathcal{C}_\beta \cap \mathcal{C}''$, for any $\beta_\beta > \beta_0$, extending $\omega_\beta = \omega|_{\mathcal{C}_\beta}$ to $\langle \mathcal{C}_\beta, \mathcal{C}'' \rangle$ amounts to extending $\omega_\beta|_{\mathcal{C}_\beta \cap \mathcal{C}''}$ to $\mathcal{C}''$ (lemma 2), ~~or~~ that is to extend ω_{β_0} from $\mathcal{C}_{\beta_0} \cap \mathcal{C}''$ to $\mathcal{C}''$. If we choose one such extension, we get up to unique isomorphism a system of extensions of the ω_β to $\langle \mathcal{C}_\beta, \mathcal{C}'' \rangle$, and by gluing them an extension of ω to $\langle \mathcal{C}', \mathcal{C}'' \rangle$.

This is all we need to conclude that $\mathcal{C}' = \mathcal{C}$.

This proof should be cleaned up. After all, we are proving that some projective system of gerbs (of the fiber functors on the $\mathcal{C}_\alpha, \alpha \in A_f$), where "projective system" is taken in a 2-categorical sense, has a non empty projective limit (again limit in a 2-categorical sense). Maybe such a translation would make the "compactness arguments" used cleaner. They were two of them: "fiber functor is a property of finite type (cf Bourb. Ex. Ch 3 §4.5)", and the noetherian property for subgroups.

The same arguments give unicity up to isomorphism of ω , over k ; we get a maximal $\mathcal{O}_\lambda$ ($\lambda \in A$) over which we have an isomorphism, and extend further if $\mathcal{O}_\lambda \neq \mathcal{O}$.

Bert

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